The neutron star has a considerable energy reservoir in its rotation, typically one rotation per second. Particles ejected from this rapidly rotating object would be caught in the rotating magnetic field and accelerated to relativistic tangential velocities. They emit high intensity bremsstrahlung, which appear as pulses of radiation with the frequency of the rotation. Such radiation pulses have been observed from cosmic sources known as *pulsars*. The discovery of a pulsar at the center of the expanding Crab nebula, which is the remnants of a supernova explosion observed by the Chinese in the year 1054 has provided strong evidence for the model outlined above. Pulsars have been suggested to be the source of the high energy cosmic ray particles observed in our atmosphere.

12.6. AGE OF THE GALAXY

Estimates of the age of our galaxy comes mainly from theoretical calculations. However, radiochemical investigations can yield information about several important steps. Thus the time since the final solidification of the earth's crust is obtained from the dating of geologic samples (Ch. 13). Nuclear clocks like U → Pb give only the age of minerals; the oldest known minerals on earth are 3.7×10^9 y.

Some glassy materials in meteorites and from the moon are assumed to be original condensed matter which never has undergone melting. The oldest ages seem all to converge at 4.6×10^9 y. Thus we conclude that this was the time when our planetary system began to form (the basis for the estimate of the age of our sun in §12.1.5).

Further information about this event has been obtained by studying tracks which nuclear decay processes leave in certain minerals (§9.1.2). Fission tracks can only persist in minerals that have not been heated because heating above 600°C erases the tracks. The fact that ^{244}Pu fission tracks have been found in iron meteorites and in lunar samples shows that ^{244}Pu existed when the planetary system formed. Because of the short half-life of ^{244}Pu (8×10^7 y) it can be concluded that such mineral samples must have formed within a few hundred million years after the nuclide ^{244}Pu itself was formed. This is probably also the time for planetary formation. The existence of primordial plutonium indicates that an r-process preceeded the formation of the planets.

Apparently, our solar system was showered by debris from a recent but distant supernova explosion which may have disturbed the "peaceful" gas cloud in our part of the universe and initiated the condensation of the solar system. At this time newly formed elements stopped being added to the solar system from the galaxy. Sufficient amounts of ^{129}I and ^{244}Pu remained after the formation of solid materials to produce characteristic isotope anomalies and fission track excesses in meteoritic and lunar materials. The spontaneous fission decay of ^{244}Pu yields ^{129}I, which decays as

$$^{129}\mathrm{I} \xrightarrow[1.6 \times 10^7 \text{ y}]{\beta^-} {}^{129}\mathrm{Xe}\ (\text{stable}) \tag{12.40}$$

The fact that meteorites containing about 100 times excess fission tracks also contain ^{129}Xe is evidence both for the existence of ^{244}Pu at the beginning of the collapse of the solar nebula, and for the rapid condensation of the meteorites from freshly produced new elements. This time has been calculated to about 80 My from the isotopic concentrations. By similar analysis we can go further back in the history of the universe and calculate the time between the formation of the elements in the galaxy and the condensation of the solar system; the result yields a time of 9.9×10^9 y (or 9.9 eons).

There are also some radiochemical evidence about the age of our Galaxy. Consider Figure 12.12, which shows a part of a nuclide chart; the zigzag arrow shows nuclides formed in the s-process, the lower right arrows nuclides formed by the r-process. ^{186}Os can only form in the s-process and ^{187}Re only in the r-process. It is likely that our solar system, and ^{187}Re, was formed from a supernova in the r-process. ^{187}Re decays ($t_{½}$ 46×10^9 y) to ^{187}Os, which also is formed from slow n-capture in ^{186}Os. The amount of ^{187}Os formed in the s-process can be calculated from estimated flux values and known cross sections (at 30 keV; 3.6×10^8 K). Thus the amount of ^{187}Os formed by decay of ^{187}Re can be calculated, and the time since the r-process determined. This rhenium-osmium clock gives us an age of 10.8 ± 2.2 eons from formation of the Galaxy until our solar system formed. If this is combined with the age of the solar system of 4.7 eons, the age of the galaxy becomes 15.5 ± 2.3 eons. Adding 1.5 eons for the time between the Big Bang and the formation of the Galaxy, the universe should have an age of 17 ± 3 eons.

This value should be compared with the most recent value of t_o (12.5), which without a correction for any change in the rate of expansion, gives 13.18 ± 0.11 eons. The disagreement between the uncorrected value, 13.2 ± 0.1 eons, and the value, 17 ± 3 eons, above indicates either that the Universe does not continue to expand at an unchanged rate, or if it does, that

(i) the expansion of the universe accelerates slowly (as indicated by resent measurements), or
(ii) Hubble's law is not valid, or
(iii) the cosmological (Einstein) constant is not unity, or
(iv) something is wrong with the Big Bang hypothesis, or
(v) the radiometric data are too uncertain.

12.7. THE EVOLUTION OF THE PLANETS AND THE EARTH

The planets are assumed to have formed out of the same cloud as, and simultaneously with, the sun. The sun contains 99.85% of the mass of the solar system, the planet Jupiter 0.1%, and all other planets 0.04%. The originally spherical cloud was disturbed and caused to rotate through some external action (e.g. supernova explosion or passage of some celestial body), the rotation leading to the formation of a disk-like cloud, and later to cloudy rings similar to those one can see around many stars and some of the planets.